Q-Tech SOC Lab – Onboarding Guide

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Network Overview

- Network: 192.168.56.0/24
- Host Machine: 192.168.56.1
- DHCP Server: 192.168.56.100 (VirtualBox)

Virtual Machines

1. Watchtower (Ubuntu Server)

- IP Address: 192.168.56.101
- Username: socanalyst
- Password: *[REDACTED]*
- Access: `ssh socanalyst@192.168.56.101`
- Purpose: Main SOC infrastructure – will host SIEM, IDS, and logging.
- Verification:
  - Run `ip a` to confirm IP is 192.168.56.101.
  - Run `ping 192.168.56.1` to test connectivity to host.

2. DevOps Workstation (Ubuntu Desktop)

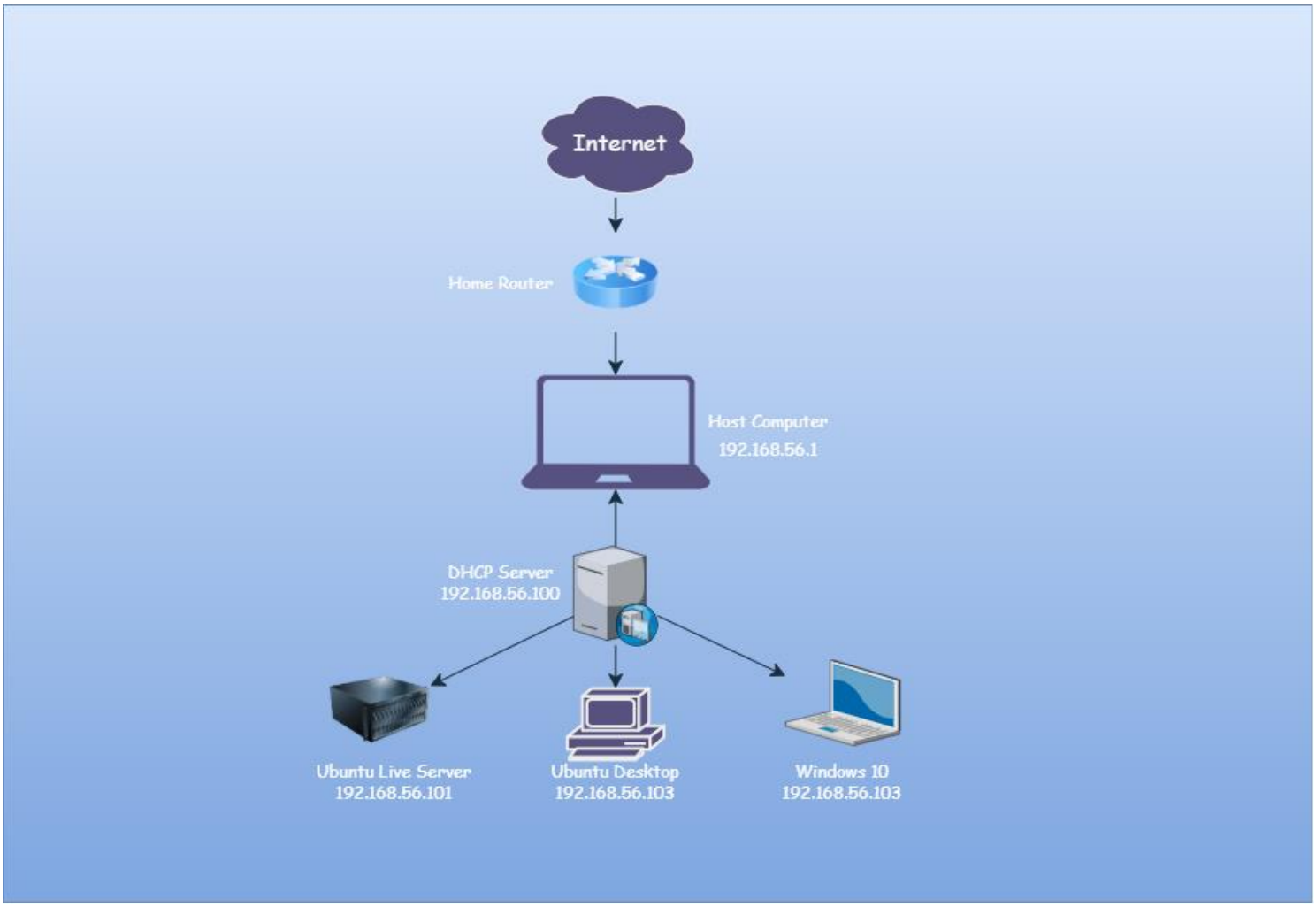
- IP Address: 192.168.56.102
- Username: devops
- Password: *[REDACTED]*
- Access: Direct console (login via VirtualBox window). SSH can be enabled la
ter.
- Purpose: Developer workstation for testing and generating traffic.
- Verification:
  - Open terminal, run `ip a` – look for `enp0s8` with IP 192.168.56.102.
  - Run `ping 192.168.56.101` to test connectivity to Watchtower.

3. Designer Workstation (Windows 10)

- IP Address: 192.168.56.103
- Username: Designer
- Password: *[REDACTED]*
- Access: Direct console (VirtualBox window).
- Purpose: Endpoint for security monitoring practice.
- Verification:
  - Open Command Prompt, run `ipconfig` – look for Host-Only adapter with IP
192.168.56.103.
  - Run `ping 192.168.56.101` to test connectivity.

Network Diagram

```
INTERNET
|
[Home Router]
|
HOST COMPUTER (192.168.56.1)
|
Host-Only Network (192.168.56.0/24)
DHCP Server: 192.168.56.100
|              |              |
Watchtower (101)   DevOps (102)   Designer (103)
```



Quick Troubleshooting

Problem	Check
Can't ping host from VM	Host firewall blocking ICMP; enable "File and Printer Sharing (Echo Request)" in Windows Firewall.
VM has wrong IP	Run `sudo dhclient -r enp0s8 && sudo dhclient enp0s8` (Linux) or `ipconfig /release && ipconfig /renew` (Windows).
SSH connection refused	On Watchtower: `sudo systemctl status ssh` – if not running, `sudo systemctl start ssh`.
No internet from VMs	Check NAT adapter is enabled in VM settings.

Verification and Documentation of IP Addresses

- **Watchtower (Ubuntu Server) – `ip a`** (relevant section):

```
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
  link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
  inet 127.0.0.1/8 scope host lo
    valid_lft forever preferred_lft forever
  inet6 ::1/128 scope host
    valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
  link/ether 08:00:27:08:27:16 brd ff:ff:ff:ff:ff:ff
  inet 10.0.2.15/24 metric 100 brd 10.0.2.255 scope global dynamic enp0s3
    valid_lft 83893sec preferred_lft 83893sec
  inet6 fd17:625c:f037:2:a00:27ff:fe08:2716/64 scope global dynamic mngtmpaddr noprefixroute
    valid_lft 86012sec preferred_lft 14012sec
  inet6 fe80::a00:27ff:fe08:2716/64 scope link
    valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
  link/ether 08:00:27:47:f4:de brd ff:ff:ff:ff:ff:ff
  inet 192.168.56.101/24 metric 100 brd 192.168.56.255 scope global dynamic enp0s8
    valid_lft 497sec preferred_lft 497sec
  inet6 fe80::a00:27ff:fe47:f4de/64 scope link
    valid_lft forever preferred_lft forever
```

- **DevOps (Ubuntu Desktop) – `ip a`**:

```
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
  link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
  inet 127.0.0.1/8 scope host lo
    valid_lft forever preferred_lft forever
  inet6 ::1/128 scope host
    valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
  link/ether 08:00:27:57:33:51 brd ff:ff:ff:ff:ff:ff
  inet 10.0.2.15/24 brd 10.0.2.255 scope global dynamic noprefixroute enp0s3
    valid_lft 83815sec preferred_lft 83815sec
```

```
    inet6 fd17:625c:f037:2:774e:2175:321d:e7c2/64 scope global temporary dynamic
      valid_lft 86146sec preferred_lft 14146sec
    inet6 fd17:625c:f037:2:9829:bcd7:8fa4:1093/64 scope global dynamic mngtmpaddr noprefixroute
      valid_lft 86146sec preferred_lft 14146sec
    inet6 fe80::89f4:e821:1e1e:6319/64 scope link noprefixroute
      valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:42:f7:3a brd ff:ff:ff:ff:ff:ff
    inet 192.168.56.102/24 brd 192.168.56.255 scope global dynamic noprefixroute enp0s8
      valid_lft 414sec preferred_lft 414sec
    inet6 fe80::55b2:9be7:64c9:fc84/64 scope link noprefixroute
      valid_lft forever preferred_lft forever
```

- **Designer (Windows 10) – `ipconfig`:**

```
Ethernet adapter Ethernet:
   Connection-specific DNS Suffix  . :
   IPv6 Address. . . . . . . . . . . : fd17:625c:f037:2:750b:ff8f:c931:23fc
   Temporary IPv6 Address. . . . . . : fd17:625c:f037:2:6908:b662:4600:95df
   Link-local IPv6 Address . . . . . : fe80::339c:8858:77cc:7036%6
   IPv4 Address. . . . . . . . . . . : 10.0.2.15
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . : fe80::2%6
                                       10.0.2.2

Ethernet adapter Ethernet 2:
   Connection-specific DNS Suffix  . :
   Link-local IPv6 Address . . . . . : fe80::28ff:5696:262a:8e63%5
   IPv4 Address. . . . . . . . . . . : 192.168.56.103
   Subnet Mask . . . . . . . . . . . : 255.255.255.0
   Default Gateway . . . . . . . . . :
```

Verification Statement: All IPs match the Q-Tech IP plan, hence I confirm the corrected IP.

Assignment 3: Simple Script to Check Connectivity

Health Check Section
Script (`check_lab_ips.sh`)

```
!/bin/bash
echo "==================================="
echo "Q-Tech IP Plan Connectivity Checker"
echo "==================================="
IPS=("192.168.56.1" "192.168.56.100" "192.168.56.101" "192.168.56.102" "192.168.56.103")
for ip in "${IPS[@]}"; do
    ping -c 2 $ip > /dev/null 2>&1
    if [ $? -eq 0 ]; then
        echo "$ip is reachable"
    else
        echo "$ip is NOT reachable"
    fi
done
```

Make the script executable by running : chmod +x check_lab_ips.sh

Execute or Run the program : sudo (bash) ./ check_lab_ips.sh

Output:

```
===================================
Q-Tech IP Plan Connectivity Checker
===================================
192.168.56.1 is NOT reachable
192.168.56.100 is reachable
192.168.56.101 is reachable
192.168.56.102 is reachable
192.168.56.103 is reachable
```

Explanation: -

`!/bin/bash` – the shebang tells the system to use the Bash interpreter to run the script.

`- > /dev/null 2>&1` redirects standard output (`> /dev/null`) and standard error (`2>&1`) to the null device, suppressing ping's normal output so only our custom messages appear.

BONUS

```
#!/bin/bash
echo "===================================="
echo "Q-Tech IP Plan Connectivity Checker"
echo "===================================="

NETWORK="192.168.56"
echo "Scanning all IPs in ${NETWORK}.0/24..."
echo ""

# Function to ping a single IP
check_ip() {
   local ip=$1
   ping -c 1 -W 1 $ip > /dev/null 2>&1
   if [ $? -eq 0 ]; then
      echo "$ip is reachable"
   fi
}

# Run pings in parallel (max 50 at a time)
for i in {1..254}; do
```

```
  ip="${NETWORK}.${i}"
  check_ip "$ip" &

  # Limit parallel jobs to 50
  if (( i % 50 == 0 )); then
    wait
  fi
done
wait

echo ""
echo "Scan complete."
```

Output:

```
====================================
Q-Tech IP Plan Connectivity Checker
====================================
Scanning all IPs in 192.168.56.0/24...

192.168.56.100 is reachable
192.168.56.103 is reachable
192.168.56.101 is reachable
192.168.56.102 is reachable

Scan complete.
```

Assignment 4: Research: Future IP Plan Elements

1. Documentation IP Ranges (RFC 5737)

The ranges `203.0.113.0/24`, `198.51.100.0/24`, and `192.0.2.0/24` are reserved by RFC 5737 for use in documentation, examples, and lab exercises. They are commonly known as TEST-NET-1, TEST-NET-2, and TEST-NET-3. These addresses will never be allocated on the public internet, making them safe for testing scenarios without risk of colliding with real systems. In our lab, they can simulate malicious external IPs for threat intelligence exercises.

2. Geo-Blocking Rationale

Organizations may block traffic from entire countries (e.g., North Korea, Russia, China) to reduce attack surface. If a company has no business operations, partners, or customers in a particular country, there is no legitimate reason for traffic from that region. Blocking such traffic reduces exposure to automated scans, exploit attempts, and commodity malware originating from those areas. However, geo-blocking is not a complete security solution; it should be combined with other controls.

3. pfSense Role

pfSense is an open-source firewall/router distribution based on FreeBSD. In our lab, it will act as the perimeter firewall, providing: - Network Address Translation (NAT) - Stateful packet filtering - VPN capabilities - Traffic shaping and monitoring - DHCP and DNS services

We will add pfSense in Week 3 to create a more realistic network boundary, with the LAN interface at `192.168.56.254` (acting as the default gateway for our VMs) and the WAN interface connected to the internet via NAT.

Assignment 5: Reflection – The Analyst's Mindset

Handling Ambiguity in a SOC

Handling ambiguity means being comfortable making decisions when you don't have all the facts. In a SOC, an alert might fire with minimal context—perhaps only an IP address and a timestamp. The analyst must decide whether to escalate based on limited information, using threat intelligence, asset value, and past experience. In my own life, I once had to decide whether to report a suspicious email to IT before I knew if it was truly a phishing attempt. I chose to report it, and it turned out to be a test from the security team. That experience taught me that it's better to act on incomplete information than to wait and risk a breach.

Why Documentation Is Often Neglected

New analysts often neglect documentation because they're focused on technical tasks and view writing as secondary. They may feel that "doing" security is more important than "writing about" security. However, poor documentation can lead to severe consequences: if an analyst is hit by a bus (or simply goes on vacation), their knowledge leaves with them. In an incident, missing documentation can delay response, cause missteps, and even lead to data loss. Documentation ensures continuity and consistency.

Peer Review Impact

The peer review exercise changed my perspective by showing me how easily instructions can be misinterpreted. My classmate couldn't find the snapshots because I had buried that information in a paragraph instead of a clear section. Conversely, my classmate's guide was so clear that I could SSH into their Watchtower in under a minute. This taught me to write for the reader, not for myself—a lesson I'll carry into my SOC career.